

CSE 350

Digital Electronics and Pulse Techniques

Assignment-04

Total – 20 marks

Question 1

05

For the NMOS given in the figure,

For M1 and M3 : $K_n = 0.3 \text{ mA/V}^2$ $V_{TN} = 0.6 \text{ V}$

For M2 : $K_n = 0.4 \text{ mA/V}^2$, $V_{TN} = 0.7 \text{ V}$

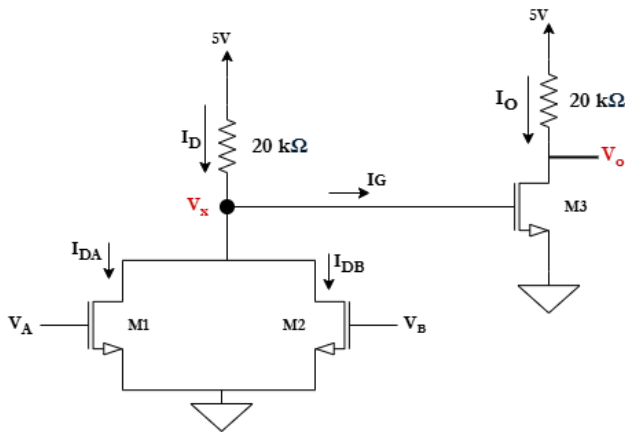


Figure 1

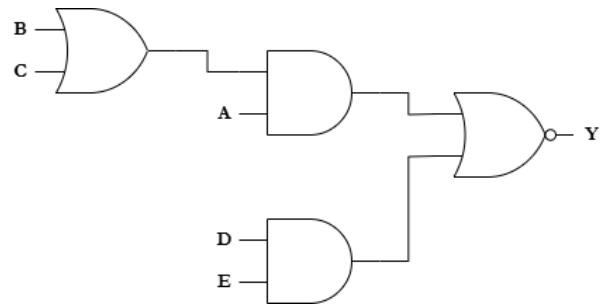


Figure 2

(a)	From the above Figure 1, determine the currents I_D , I_{DA} , I_{DB} , I_O , V_X and V_O for the cases both inputs are high and both inputs are low. (Consider 5 V as logic high and 0 V as logic low)	2
(b)	Determine the power dissipations for the cases both inputs are high and both inputs are low.	1
(c)	Find the Boolean expression for output Y from Figure 2 and using that expression design an equivalent CMOS logic circuit of Figure 2.	2

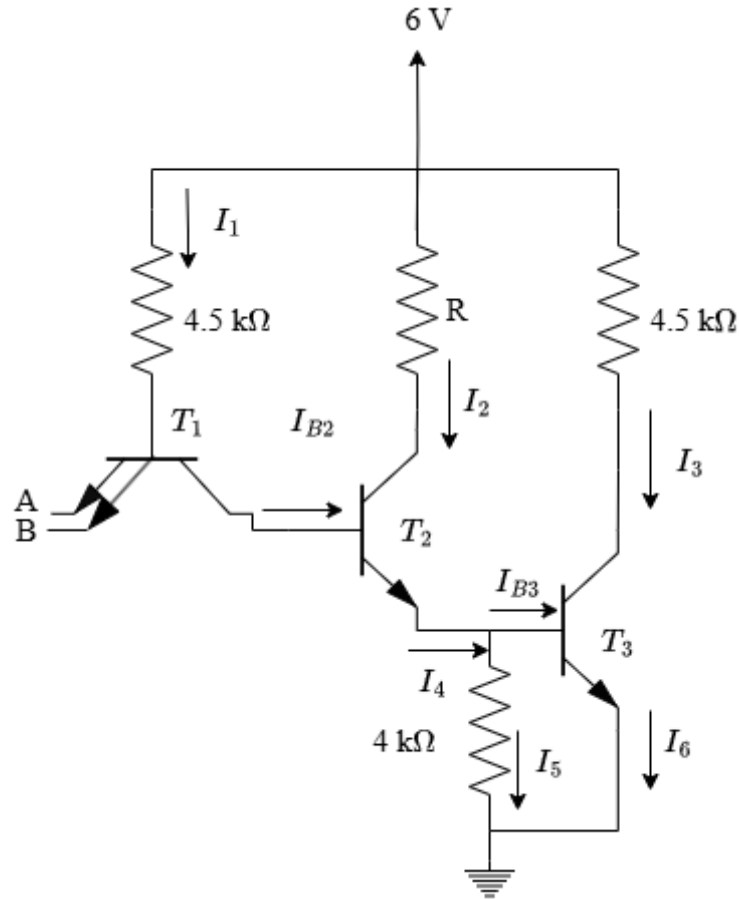
Question 2

05

For the given TTL NAND gate,

$\beta_R = 0.2$, $V_{CE}(\text{saturation}) = 0.2 \text{ V}$, $V_{BE}(\text{saturation}) = 0.8 \text{ V}$, $V_{BC}(\text{reverse active}) = 0.7 \text{ V}$

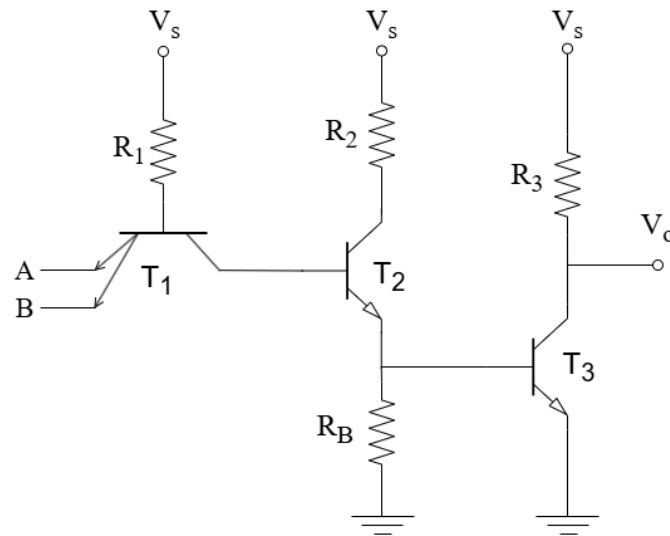
$V_{in}(\text{high}) = 6 \text{ V}$, $V_{in}(\text{low}) = 0.2 \text{ V}$



(a)	If $I_6 = 4.74 \text{ mA}$ when both inputs are high , find the value of R .	[5]
-----	--	-----

Question 3

05

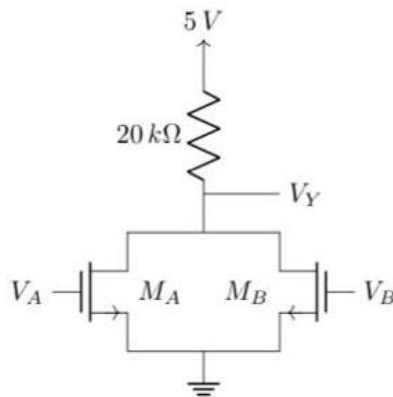


$V_s = 9\text{ V}$, $R_2 = 2\text{ k}$, $R_3 = 4.4\text{ k}$, $R_B = 4.4\text{ k}$

(a)	Determine the power dissipation of this TTL circuit when both inputs are HIGH. You must show your calculation. Assume $R_1 = 6.7\text{ k}$ for this question only	3
(b)	Suppose you are selling this TTL logic gate to a customer and he states a strict criteria of keeping the power dissipation below 1.23 mW when both inputs are LOW (0 V). Determine the maximum value of R_1 you can use to satisfy this criteria.	2

Question 4

05



For the NOR gate shown, $V_{TN_A} = 0.25\text{ V}$, $V_{TN_B} = 0.3\text{ V}$, $K_{n_A} = 0.4 \frac{\text{A}}{\text{V}^2}$, $K_{n_B} = 0.5 \frac{\text{A}}{\text{V}^2}$.

(a)	Find the lower threshold of output voltage.
(b)	Find the average power dissipation.
(c)	If a new NMOS M_C is used instead of M_B such that the drain currents of M_A & M_C are equal when both inputs are high, find the value of K_{n_C} . It is given that $V_{TN_C} = V_{TN_B} = 0.3\text{ V}$.